

Benjamin Steenhoek, PhD

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AI/ML researcher building code-editing and agentic systems for software engineering. Current focus areas are LLM post-training and agent harness engineering. Research interests include agent evaluation, LLM fine-tuning, software security, and program analysis. Based in Des Moines, Iowa.

EXPERIENCE

Senior Researcher

Fall 2024–Present

[Microsoft Code|AI](#) · Des Moines, IA

- Drove research engineering for **GitHub Copilot's primary agentic evaluation platform**, building the scalable harness environment and tooling and onboarding key benchmarks that supported its growth from its **greenfield prototype stage to 1M+ monthly executions**, powering evaluations for flagship coding agents including [VS Code Agent](#), [Copilot Cloud Agent](#), and [Copilot CLI](#).
- Led training, evaluation, and rollout of customizable [GitHub Copilot Inline Suggestions](#), **addressing 10+ community issues and increasing accepted suggestions by 10%**.

Research Intern

May 2023–Aug 2024

[Microsoft Code|AI](#) · Des Moines, IA

- Conducted a user study on in-IDE AI-powered security vulnerability detection & repair with **17 professional developers across 24 projects, 6.9K files, and 1.7M+ lines of code**; [Closing the Gap, ICSE 2025](#) (21% acceptance rate).
- **Improved unit-test readability and maintainability by up to 21%** through reinforcement learning; the fine-tuned Codex model was **less than half the cost of GPT-4 and outperformed it on 4 of 7 quality metrics**; [Reinforcement Learning from Automatic Feedback, DeepTest @ ICSE 2025](#).

Research Assistant

2020–2024

[Program Analysis & AI Lab](#) · Iowa State University · Ames, IA

- Achieved **state-of-the-art Big-Vul vulnerability detection performance (96.46 F1)** by combining a dataflow-inspired graph model with an LLM. Our model trained in 9 minutes—**75× faster than the strongest baseline**—and detected 8.7 of 17 real-world vulnerabilities where baselines found none; [DeepDFA, ICSE 2024](#) (code; 22% acceptance rate).
- Built a [C/C++ execution tracing engine](#) from scratch for [TRACED, ICSE 2024](#) (code; 22% acceptance rate), enabling execution-aware pre-training that **improved complete path prediction by 12.4% and runtime value prediction by 25.2%**. Built a Java counterpart later used for [CodeSense, ICLR 2026](#).
- Reproduced and evaluated **9 state-of-the-art vulnerability detection models across 2 datasets**, uncovering substantial run-to-run variance and low agreement among model outputs; [An Empirical Study of Deep Learning Models for Vulnerability Detection, ICSE 2023](#) (code; 26% acceptance rate).
- Collaborated with Columbia University's ARiSE Lab and Carnegie Mellon University's CERT division; built open-source static and dynamic analysis tools including [tree-climber](#) and [pal-tools](#).

SELECTED PUBLICATIONS

See all: 🎓 scholar.google.com/citations?user=uJk56S4AAAAJ

1. Monoshi Kumar Roy, Simin Chen, **Benjamin Steenhoek**, Jinjun Peng, Gail Kaiser, Baishakhi Ray, and Wei Le. (2026). *CodeSense: a Real-World Benchmark and Dataset for Code Semantic Reasoning*. ICLR'26. [Paper](#) · [Code](#)
2. Spandan Garg, **Benjamin Steenhoek**, and Yufan Huang. (2026). *Saving SWE-Bench: A Benchmark Mutation Approach for Realistic Agent Evaluation*. CAIN'26. [Paper](#) · [Code](#)
3. **Benjamin Steenhoek**, Kalpathy Sivaraman, Renata Saldivar Gonzalez, Yevhen Mohylevskyy, Roshanak Zilouchian Moghaddam, and Wei Le. (2025). *Closing the Gap: A User Study on the Real-world Usefulness of AI-powered Vulnerability Detection & Repair in the IDE*. ICSE'25. [Paper](#)
4. **Benjamin Steenhoek**, Michele Tufano, Neel Sundaresan, and Alexey Svyatkovskiy. (2025). *Reinforcement Learning from Automatic Feedback for High-Quality Unit Test Generation*. DeepTest @ ICSE'25. [Paper](#)
5. **Benjamin Steenhoek**, Md Mahbubur Rahman, Monoshi Kumar Roy, Mirza Sanjida Alam, Hengbo Tong, Swarna Das, Earl T. Barr, and Wei Le. (2024). *To Err is Machine: Vulnerability Detection Challenges LLM Reasoning*. arXiv. [Paper](#)
6. Yanguibo Ding, **Benjamin Steenhoek**, Kexin Pei, Gail Kaiser, Wei Le, and Baishakhi Ray. (2024). *TRACED: Execution-aware Pre-training for Source Code*. ICSE'24. [Paper](#) · [Code](#)
7. **Benjamin Steenhoek**, Hongyang Gao, and Wei Le. (2024). *Dataflow Analysis-Inspired Deep Learning for Efficient Vulnerability Detection*. ICSE'24. [Paper](#) · [Code](#)
8. **Benjamin Steenhoek**, Md Mahbubur Rahman, Shaila Sharmin, and Wei Le. (2023). *Do Language Models Learn Semantics of Code? A Case Study in Vulnerability Detection*. arXiv. [Paper](#)
9. **Benjamin Steenhoek**, Md Mahbubur Rahman, Richard Jiles, and Wei Le. (2023). *An Empirical Study of Deep Learning Models for Vulnerability Detection*. ICSE'23. [Paper](#) · [Code](#)

10. Ashwin Kallingal Joshy, Xueyuan Chen, Benjamin Steenhoek, and Wei Le. (2021). *Validating Static Warnings via Testing Code Fragments*. *ISSTA'21*. [Paper](#)

EDUCATION

PhD, Computer Science	Iowa State University	2019–2024
Dissertation: Understanding and improving deep learning models for vulnerability detection		
MS, Computer Science	Iowa State University · GPA 3.91/4.00	2019–2021
Thesis: Refactoring programs to improve the performance of deep learning for vulnerability detection		
BS, Computer Science	Bob Jones University · magna cum laude · GPA 3.84/4.00	2016–2019

INVITED TALKS & SERVICE

- Reviewer · [ACM TOSEM](#) 2025, 2026
- Reviewer · [Empirical Software Engineering](#) 2024, 2026
- Program Committee · [SVM](#) 2026
- Program Committee · [FSE'25 IVR](#) 2025
- Program Committee · [FORGE](#) 2025
- Invited Talk · [IEEE DISTILL](#) 2025
- Reviewer · [IEEE TIFS](#) 2023

SELECTED PROJECTS

See all: github.com/bstee615

- [DeepDFA](#) — efficient, dataflow-inspired vulnerability detection
- [TRACED](#) — C/C++ execution tracing for model pre-training
- [cfactor](#) — policy-driven refactoring for C programs
- [tree-climber](#) — program analysis tools for C built on tree-sitter
- [pal-tools](#) — dynamic analysis and code-generation utilities
- [rrun](#) — Git-aware remote command runner over SSH
- [wslwatch](#) — watchdog that keeps WSL2 distributions running